SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:50**

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Nalluri Charan Sai (192372003) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N Charan Sai

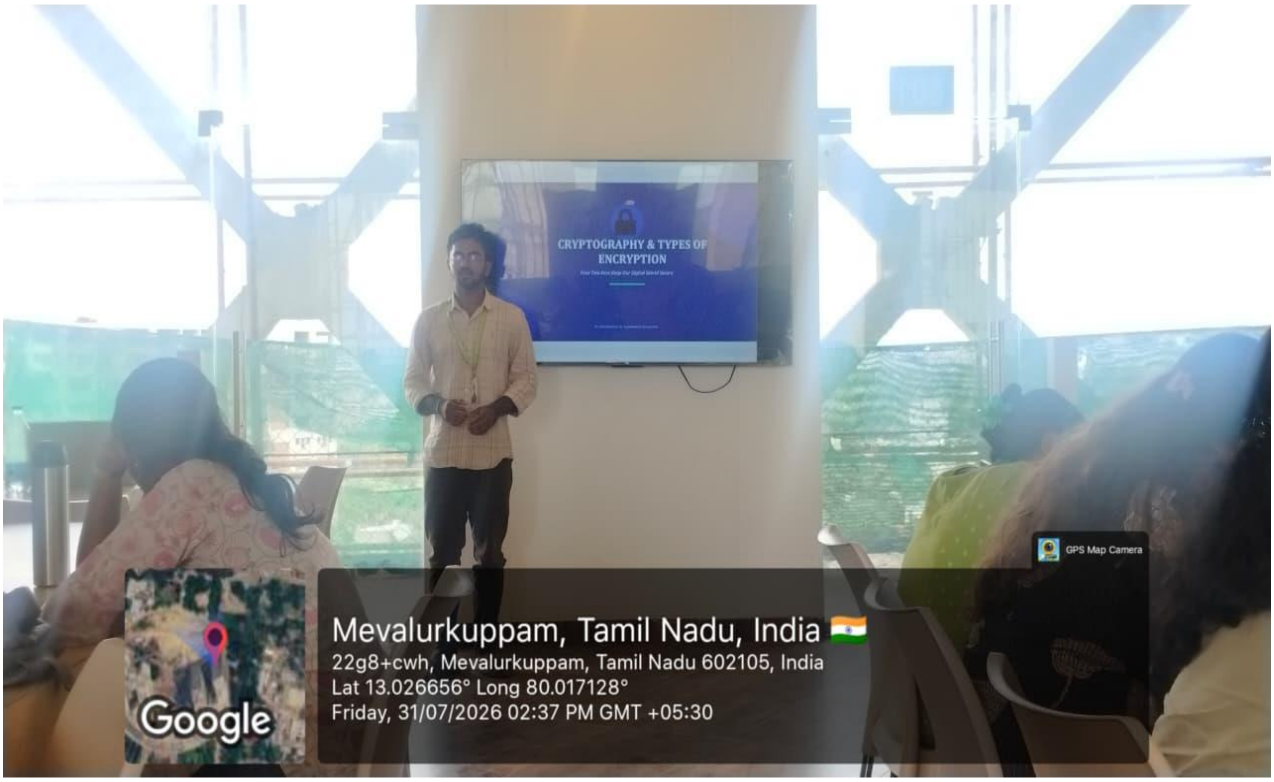
Annexure A

TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers"
2. "Substitution vs Transposition: Which Provides Better Security?"
3. "Security Attacks, Threats, Risks and Their Impact on Security Goals"
4. "Steganography vs Cryptography: Hidden Security in the Digital World"
5. "Role of Number Theory in Cryptography: From Euclid to Euler"

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation


CRYPTOGRAPHY & TYPES OF ENCRYPTION
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CRYPTOGRAPHY & TYPES OF ENCRYPTION

How Two Keys Keep Our Digital World Secure

An Introduction to Asymmetric Encryption

What Is Cryptography?

Cryptography is the science of protecting information by converting it into a secret code so that only authorized people can read it.

The word comes from:

Crypto = Hidden or Secret

Graphy = Writing

Plaintext

Only the right people can read the message.

Encryption

Converting the message into a secret code.

Decryption

Converting the secret code back into the original message.

How Public-Key Cryptography Works

Every user has a key pair: a public key anyone can see, and a private key only they hold.

1. Generate Keys

A public key and a matching private key are created together.

2. Share the Public Key

The public key is sent openly — anyone can use it.

3. Encrypt the Message

The sender locks the message using the receiver's public key.

4. Decrypt with Private Key

Only the matching private key can unlock and read it.

Symmetric vs. Asymmetric

Two families of encryption, built for different jobs.

Symmetric Encryption

Symmetric encryption uses the same key for encryption and decryption.

The sender uses a secret key to convert the message into a coded form.

The receiver uses the same key to get the original message back.

It is fast and commonly used for protecting large amounts of data.

Example: AES (Advanced Encryption Standard)

Asymmetric Encryption

Asymmetric encryption uses two keys: a public key and a private key.

The public key is used for encryption, and the private key is used for decryption.

The private key is kept secret by the owner.

It provides better security for communication over the internet.

Real-World Applications

Public-key cryptography runs quietly behind everyday technology.

HTTPS & Web Browsing

Secures the padlock icon in your browser and protects sites you visit.

Secure Email (PGP)

Encrypts messages so only the intended reader can open them.

SSH & Remote Access

Lets developers log into servers without sending a password.

Blockchain & Crypto Wallets

Signs transactions to prove ownership without revealing private keys.

Thank you!